

HOUSE PRICE PREDICTION

using Linear Regression

Regression Model for Prediction

Internship: CodeOrbit Tech

Role: Machine Learning Intern

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Project: 3 — Regression Model for Prediction

1. Project Overview

This project focuses on predicting house values using a Linear Regression model. The California Housing dataset is used for the regression task. The workflow covers data inspection, train-test splitting, Linear Regression, prediction, model evaluation, and visualization of actual versus predicted values.

2. Objective

The objectives are to select a regression dataset, prepare the data, split it into training and testing sets, build a Linear Regression model, generate predictions, evaluate the model, and visualize actual values against predicted values.

3. Dataset

The California Housing dataset contains 20,640 records and 8 input features. The target variable is **MedHouseVal**, representing the median house value. The eight input features are MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude.

4. Data Preprocessing

The dataset was loaded using Scikit-learn and inspected for missing values, duplicate records, dataset structure, features, and the target variable. The target variable **MedHouseVal** was separated from the input features before model training.

5. Train-Test Split

The dataset was divided using an 80/20 train-test split. 16,512 records were used for training and 4,128 records were used for testing.

6. Model

A Linear Regression model was trained using the eight housing-related features. The trained model was then used to predict house values for the unseen test data.

7. Model Evaluation

The model was evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² Score. These metrics provide different views of prediction error and explained variation.

8. Final Results

Metric	Result
Training Records	16,512
Testing Records	4,128
Number of Features	8
MAE	0.5332

MSE	0.5559
RMSE	0.7456
R ² Score	0.5758

The model achieved an R² Score of 0.5758, meaning that approximately 57.58% of the variation in the target variable was explained by the model.

9. Visualization

The notebook includes an Actual vs Predicted House Values scatter plot. Points closer to the diagonal reference line indicate predictions that are closer to the actual values.

The actual-vs-predicted visualization is available in the submitted notebook.

10. Key Insights

- The model was successfully trained on 16,512 records and tested on 4,128 records.
- Eight housing-related features were used as predictors.
- The R^2 Score was 0.5758, indicating that the model explains approximately 57.58% of target variation.
- The Actual vs Predicted visualization provides a direct view of prediction performance.
- The Linear Regression model provides a useful baseline, while further improvement remains possible.

11. Conclusion

This project demonstrated the use of Linear Regression for predicting house values. The complete workflow included dataset loading, data inspection, feature-target separation, train-test splitting, model training, prediction, evaluation, and visualization. The model achieved an R^2 Score of 0.5758 and provided practical experience with regression modeling and machine learning evaluation techniques.

12. Future Scope

- Try other regression algorithms.
- Apply feature engineering techniques.
- Tune model parameters.
- Compare Linear Regression with other regression models.
- Use additional relevant features.

13. Technologies Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Google Colab, and Jupyter Notebook.